# Architecture & Systems Engineering

*Week 3: System Architecture*

## Project Portfolio

Name

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# Week 3 Project

## Overview

In the third project activity of this course, your team will build on a specific project of your own choosing – such as the last system you worked on, or a system from a previous project. Please ensure that the system has at least a medium (but preferably high) level of complexity -- such as a car, satellite, an enterprise server, or an open-source software.

Note that some Scratch Pages are included at the end of the project document for you to capture any ideas, sketches, etc. that you may have as you work through the project. These will not be assessed and you are not required to submit them with your project (but you may do so if you think they offer any additional insight into your thinking process!).

**REQUIRED STEPS:**

**Step 1**: Select your system for this project.

**Step 2**: Produce a solution-neutral function and concept.

**Step 3**: Develop and analyze a set of architectural decisions.

**Step 4**: Review and submit your project.

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# STEP 1: SYSTEM SELECTION

*Provide a diagram for a system with medium to high level of complexity for your project. Examples of suitable systems include a car, a satellite, an enterprise server, or an open-source software program. Additionally, write 1-2 sentences that summarize the high level stakeholder needs for your system. If you are working in a team, ensure that (1) at least one team member is knowledgeable about the system and (2) all members feel comfortable analyzing it together.*

*Please remember the file size limit and resize or paste the image URL instead, as needed.*

**Name of System:** Pulsar - Electromagnetic warfare

**System Image/Diagram/Schematic View of Form and Stakeholder Needs Summary**

https://www.anduril.com/pulsar

Military operators need an electromagnetic warfare system that can detect, classify, and counter diverse electronic threats across the spectrum. Adaptability must be in hours to days rather than months.

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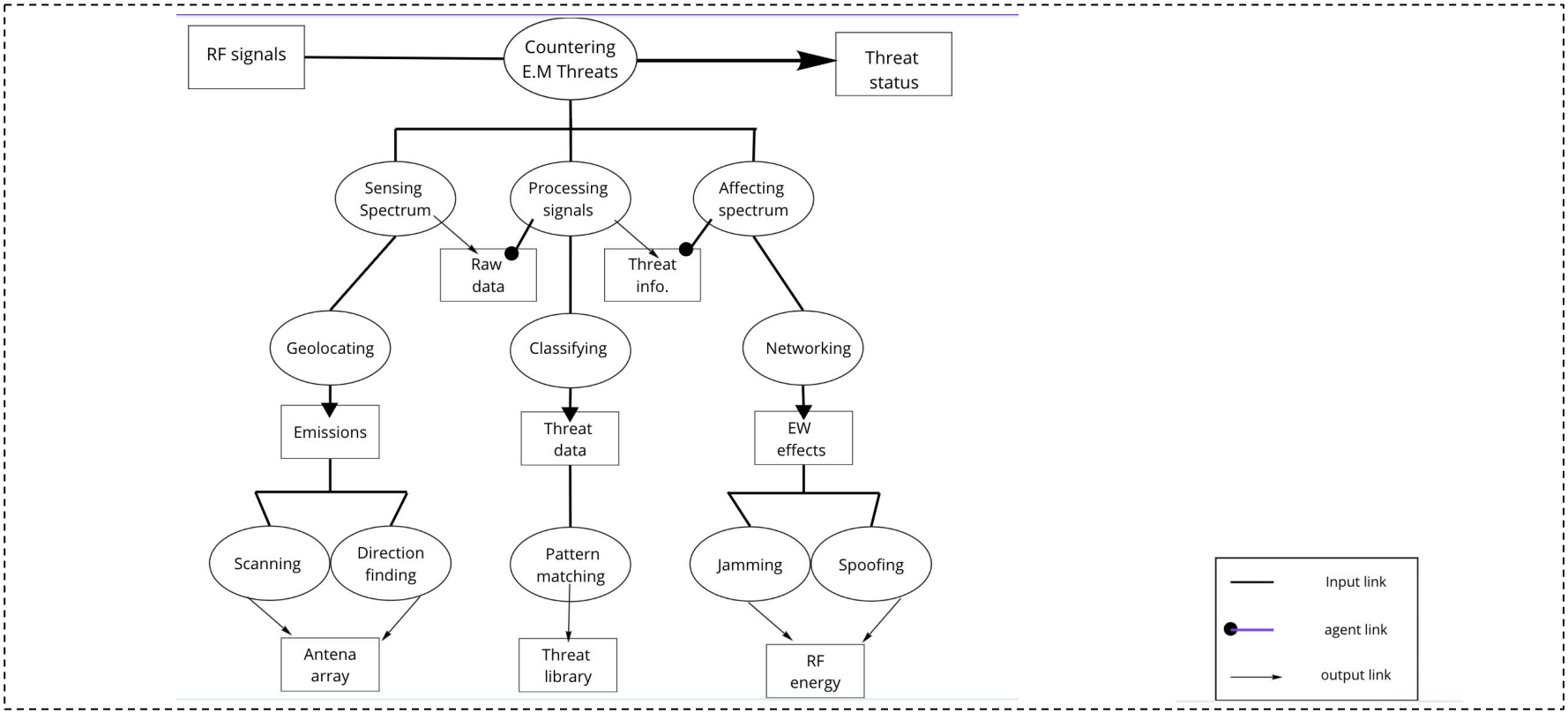
# STEP 2A: SOLUTION-NEUTRAL FUNCTIONS

*For your system, provide a diagram showing a solution neutral (eg. "accessing") to solution specific (eg. "pushing") hierarchy. This is a process diagram. Per course examples, decomposition begins with a top-level starting at zero. You should include three additional levels, up to and including level 3. In your diagram, include operands (e.g. cork, bottle, wine) but not forms associated with each process. If needed, refer to the Wine Bottle System example in the "Identifying the Solution-Neutral Function" unit.*
*Please remember the file size limit and resize or paste the image URL instead, as needed*

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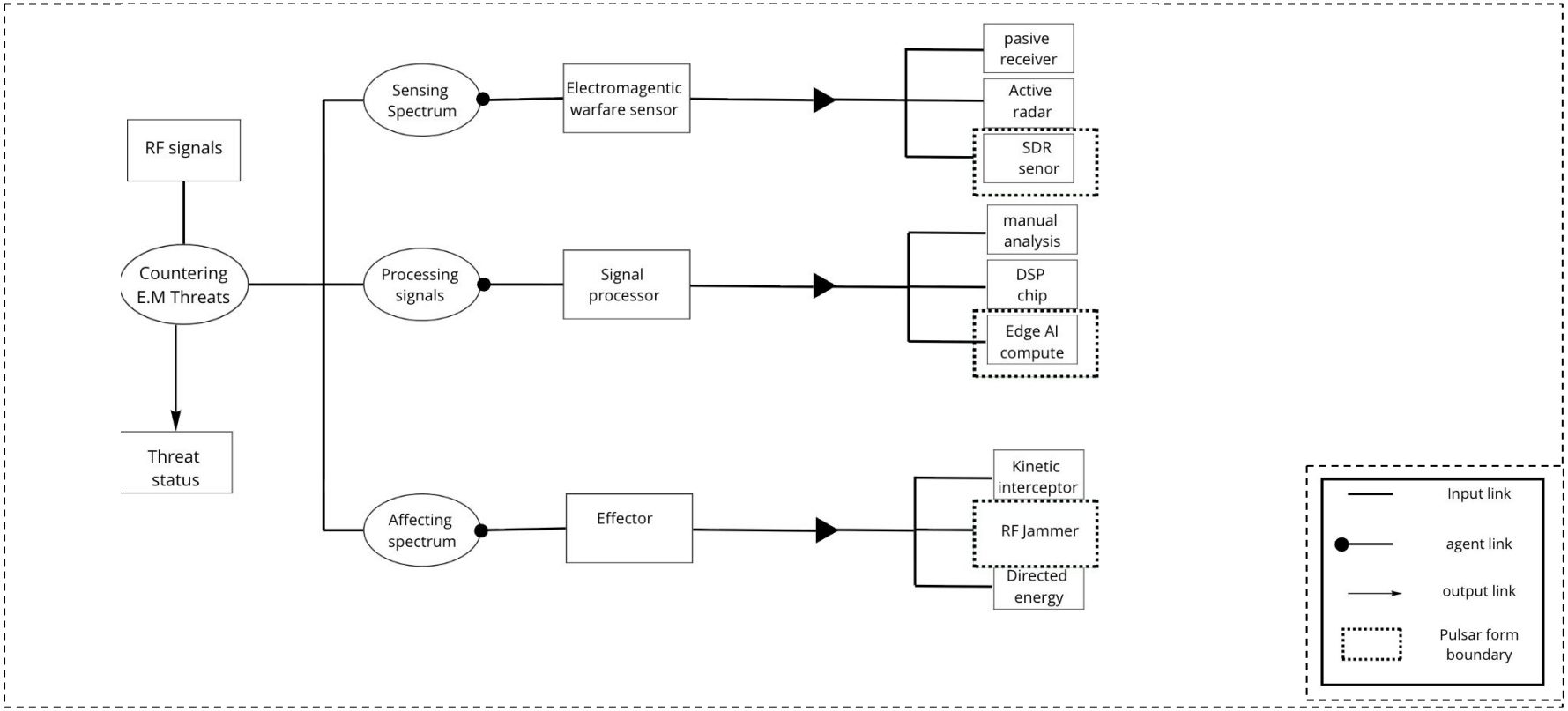
# STEP 2B: FORMS

*For the solution-neutral functions produced in step 2A, develop a set of possible forms that could be used. For example, in the Wine Bottle Example if you chose "Cork Removing", now illustrate several forms by which you could accomplish this – Cork Suctioning Using a Vacuum Device, Cork Pulling Using a Corkscrew, etc. Mark your selected concept by drawing a system boundary around the selected function and form.*

*Please remember the file size limit and resize or paste the image URL instead, as needed.*

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# STEP 3A: ARCHITECTURAL DECISIONS

*Develop a set of architectural decisions (minimum of 5) for your system. Keep in mind that these should be impactful decisions that are connected and sensitive to key stakeholder metrics. Provide a rationale for the decision and two alternative options.*

| Architectural decision | Rationale | Option(s) |
|---|---|---|
| SDR architecture | enables rapid reprogramming to counter new threats. | FPGA based radio or fixed-frequency receiver. |
| modular multimission design | single platform supports sensing, jamming and C2 simultaneously | separated stacked systems. |
| tripod-mounted fixed | 360 deg. coverage | vehicle mounted only. handled jammer. |
| lattice mesh networking | distributed learning and classifier. | standalone operation |
| open architecture | third party software integration accelerates capability development. | propietary closed system or government-only development |
| Under 4 kW power draw | Operable from standard generators without dedicated power infrastructure. | high-power shelter system or battery-only operations. |
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# STEP 3B: ARCHITECTURAL DECISIONS

*Classify each architectural decision from Step 3A as connected and sensitive in the below matrix. However, do **not** automatically classify all decisions as highly connected and sensitive. Even though architectural decisions should be both connected and sensitive, it is still possible to actively order them.*

| | **Less Connected** | **More Connected** |
|---|---|---|
| **More Sensitive** | tripod-mounted fixed<br>open architecture | SDR architecture<br>modular multimission design<br>lattice mesh networking |
| **Less Sensitive** | Under 4 kW power draw | |

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# STEP 3C: ARCHITECTURAL DECISIONS

*For the set of architectural decisions developed, include a small write-up with constructive feedback that touches on the following. Do these decisions represent the architecture of the system? Are they true architectural decisions?*

*Please remember the file size limit and resize or paste the image URL instead, as needed.*

SDR architecture, modular multimission design, and Lattice mesh networking are highly coupled and sensitive. SDR shapes the entire processing pipeline and software architecture, switching to fixed-frequency would require redesigning the threat library and signal chain. Modular design couples sensing, jamming, so changing one affects the others. Lattice mesh connects all nodes and enables distributed learning. Without it, the system loses its collective intelligence advantage.

Tripod-mounted form factor and open architecture are highly coupled but less sensitive. The tripod determines antenna layout and sealing requirements, but doesn't directly affect detection or jamming performance. Open architecture defines third-party integration but doesn't change core EW metrics.

Under 4 kW power draw is loosely coupled and low sensitivity. Power can be adjusted independently of the SDR, AI, or networking subsystems and has limited impact on threat response time or detection range. It is not a true architectural driver.

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